

A Basis for the 2-Kernel of Euler's Totient

Exact finite-level ranks and consequences for Erdős Problem #249

WILL COOK*

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ABSTRACT

For every $e \geq 1$, the dyadic sections of Euler's totient through level e have rational rank $2^e + 1$. The two zero-residue sections and the odd-residue sections form a basis, and two families of scalar reductions generate every relation. Divisibility makes an evaluation matrix diagonal modulo a prime, with nonzero diagonal entries. The construction gives rank $k^e + 1$ in every base $k \geq 2$ and an integral normal form. Separately, $\sum_{n \geq 1} (\varphi(n) \bmod m) 2^{-n}$ is irrational for every $m \geq 3$, with a complete rationality classification at dyadic moduli. For $S = \sum_{n \geq 1} \varphi(n) 2^{-n}$, one nonintegral tail shift forces full-depth certificates in every sufficiently late pair of multipliers. Irrationality is equivalent to an explicit central-residue gap for every prospective denominator; its arithmetic supply remains open.

1 A basis and all its relations

Set $\varphi(0) = 0$. For $k \geq 2$, the k -kernel consists of the sequences $n \mapsto \varphi(k^j n + r)$ with $j \geq 0$ and $0 \leq r < k^j$. The following theorem gives a unique coordinate system at every finite level.

Theorem 1.1 (exact rank of every totient k -kernel truncation). *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{ F_{0,0}^{(k)}, F_{1,0}^{(k)} \} \cup \{ F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r \}$$

is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1} F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p} \right).$$

The retained condition is $k \nmid r$. At $k = 6$, for example, the residue $r = 2$ is retained. At base 2 and level 3, the fifteen sections have nine coordinates; the omitted positive-residue sections satisfy

$$\varphi(8n + 2) = \varphi(4n + 1), \quad \varphi(8n + 4) = 2\varphi(2n + 1), \quad \varphi(8n + 6) = \varphi(4n + 3).$$

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1.1 The evaluation matrix

The two ingredients are scalar reduction and arithmetic independence. The local formula for φ gives the displayed reductions. For a prime $p \mid k$ that does not divide u , the argument $k^{j-t}n + u$ is a p -adic unit, so multiplication by k^t contributes its power of p and one factor $1 - 1/p$. When $p \mid u$, the argument already contains p , and only its power contributes. This gives $C_k(t, u)$.

For independence, consider finitely many forms $L_i(n) = a_i n + b_i$ with positive slopes and $a_i b_j \neq a_j b_i$ for $i \neq j$. A finite shift makes the intercepts positive. Write $L_i = g_i A_i$, where A_i is primitive. Choose an odd prime ℓ avoiding the slopes, all $g_i \varphi(g_i)$, and the nonzero cross determinants. In row i , choose distinct primes $q_{ij} \equiv 1 \pmod{\ell}$ for $j \neq i$, outside the same finite exceptional set, and impose

$$L_j(n) \equiv 0 \pmod{q_{ij}}, \quad A_i(n) \equiv 2 \pmod{\ell}.$$

The Chinese remainder theorem combines these congruences. The resulting progression for $A_i(n)$ is reduced: a prime q_{ij} dividing $A_i(n)$ would also divide the nonzero cross determinant; ℓ is excluded by the chosen residue; and primitivity excludes the primes of the slope. Dirichlet's theorem supplies arbitrarily large primes $p = A_i(n)$ with $p \nmid g_i$. Only the selected affine form is made prime in this row. The diagonal value is $\varphi(g_i)(p-1) \not\equiv 0 \pmod{\ell}$. Each off-diagonal value is divisible by ℓ , since $q_{ij}-1$ divides the corresponding totient. One evaluation per row gives

$$(\varphi(L_j(n_i)))_{i,j} \equiv \text{diag}(u_1, \dots, u_s) \pmod{\ell}, \quad u_i \neq 0.$$

The determinant is nonzero over $\mathbb{Q}$.

Apply this construction after restricting a relation among the retained sections to $n = km + 1$. The positive-residue sections give pairwise nonproportional forms

$$k^{j+1}m + (k^j + r), \quad k \nmid r,$$

and the remaining form is $km + 1$. Equality between two affine fractions would make the residue at the larger level divisible by k . The matrix therefore eliminates every positive-residue coefficient. If A, B are the two zero-residue coefficients, the restriction gives $A + B\varphi(k) = 0$. Evaluation at $n = k$ gives $A + Bk = 0$, since $\varphi(k^2) = k\varphi(k)$. As $\varphi(k) < k$, both coefficients vanish. The reductions give spanning and the count is

$$2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1.$$

This proves Theorem 1.1. The scalar reduction at level $j - t$, with the Euler product written out, is [Lean source](#).

Corollary 1.2 (Dyadic basis). *The family*

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.

Proof. Every finite subfamily lies in a sufficiently high truncation. The theorem gives its independence; the reductions give equality of the full spans. The level-zero clause is [Lean source](#). $\square$

Corollary 1.3 (Integral normal form). *The canonical family is also a $\mathbb{Z}$ -basis of the module generated by the sections through level e . Its integral relation module has a basis with one elementary reduction per omitted channel and rank $\sum_{j=1}^{e-1} k^j$.*

Proof. The denominator $\prod_{p|k, p \nmid u} p$ in $C_k(t, u)$ divides k , so all reduction scalars are integers. In the free module on the channels, subtracting one elementary reduction for each omitted coordinate leaves a relation among the independent retained coordinates. The omitted-coordinate pivots are all one, which also gives independence of the elementary relations. The coordinate module carries an [integral basis](#) in Lean. The basis of the relation module stays outside the Lean development. $\square$

The module here is the one generated by the channels. It is not saturated in the group of all integer-valued sequences: at base 2 and $e \geq 2$, $\varphi(4n+3)/2$ is integer-valued but has canonical coefficient $1/2$.

1.2 A reusable consequence

Corollary 1.4 (Freezing periodic coefficients). *If L_i are pairwise nonproportional positive-slope affine forms and w_i are rational-valued periodic sequences, then*

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \quad \implies \quad w_i(n) = 0 \quad \text{for every } i, n.$$

Proof. Choose a common period Q . On each progression $n = a + Qt$, the coefficients are constant and the cross determinants become $Q(a_i b_j - a_j b_i) \neq 0$. Shift beyond the threshold and apply the affine independence just proved. The statement is [Lean source](#), with a separate period for each coefficient. $\square$

The same argument preserves the basis after deletion of a finite prefix. For the two zero channels, large $n \equiv 1 \pmod{k}$ and large powers of k replace the two evaluations used above.

2 Bounded residues and rationality

The residue theorem uses the same separation principle: force neighbouring coefficients to vanish while retaining one central value.

Theorem 2.1. *For every $m \geq 3$,*

$$A_m = \sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$, the series $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k) 2^{-n}$ is rational exactly when f is constant on the even residue classes. If that constant is c , its value is $3f(1)/4 + c/4$.

Lemma 2.2 (A bounded isolated pulse). *Let $a_n \in \mathbb{Z}$ satisfy $|a_n| \leq C$. Suppose that for arbitrarily large L there is $N > L$ such that $a_N \neq 0$ and $a_{N+t} = 0$ for $0 < |t| \leq L$. Then $\sum_{n \geq 1} a_n 2^{-n}$ is irrational.*

Proof. If the sum has denominator q , every scaled tail $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$ belongs to $q^{-1}\mathbb{Z}$. Choose L so large that $C2^{-L} < 1/q$. The right zero block gives $|T_N| \leq C2^{-L} < 1/q$, hence $T_N = 0$. The left zero block now gives $T_{N-L-1} = a_N 2^{-L-1}$, a nonzero element of $q^{-1}\mathbb{Z}$ with absolute value less than $1/q$, a contradiction. The statement is [Lean source](#). $\square$

Proof of Theorem 2.1. Fix a unit $s \pmod{m}$ and a block length L . For each nonzero $t \in [-L, L]$, choose a distinct prime $q_t \equiv 1 \pmod{m}$ larger than L and outside the prime divisors of m . Prescribe

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t}.$$

This is a reduced progression, because each $q_t > |t|$. Dirichlet supplies arbitrarily large primes p in it. Then $m \mid \varphi(p+t)$ for every $0 < |t| \leq L$, whereas $\varphi(p) \equiv s-1 \pmod{m}$.

Subtract $f(0)$ from the observable and clear its denominators. Whenever $f(s-1) \neq f(0)$, the preceding construction gives arbitrarily long bounded isolated pulses, so Lemma 2.2 applies. For the least-residue map choose $s = -1$: its central residue is $m-2 \neq 0$ when $m \geq 3$. For $m = 2^k$, every even residue r has $r+1$ a unit, so a non-constant restriction to the even residues supplies such a pulse. Conversely, $\varphi(n)$ is even for every $n \geq 3$. Constancy on the even classes therefore makes all terms after $n = 2$ constant, giving the stated rational value. This converse requires no recurrent-support theorem. The three clauses together are [Lean source](#). $\square$

The bounded-pulse proof fixes its coefficient bound before choosing the block. For S , the termwise condition $2^L \mid \varphi(N+L)$, with $N+L > 1$, forces $2^L \leq \varphi(N+L) \leq N+L-1$. Hence the tail envelope $(N+L+2)2^{-L} > 1$ cannot exclude any positive denominator. The following criterion instead measures cancellation in a weighted prefix.

3 The actual series and the remaining residue gap

Write

$$S = \sum_{n \geq 1} \varphi(n) 2^{-n}, \quad R_N = \sum_{j \geq 1} \varphi(N+j) 2^{-j}, \quad \Delta_h(N) = R_{N+h} - R_N.$$

The prefix identity $2^N S = \Phi_N + R_N$, with $\Phi_N = \sum_{n=1}^N \varphi(n) 2^{N-n} \in \mathbb{Z}$, gives

$$\Delta_h(N) \equiv 2^N (2^h - 1) S \pmod{\mathbb{Z}}. \tag{1}$$

In particular a nonintegral tail difference is a precise arithmetic statement about S .

3.1 One seed removes the full-depth restriction

For $h, N, L \in \mathbb{N}$, let

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j)) 2^{L-1-j}, \quad B = N + h + L + 2.$$

Write $K(h, N, L)$ for

$$B < D_{h,N,L} \bmod 2^L < 2^L - B.$$

The exact truncation and $\varphi(n) \leq n$ give

$$|2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (2)$$

Thus some L satisfies $K(h, N, L)$ exactly when $\Delta_h(N) \notin \mathbb{Z}$: necessity follows from the strict inequalities; sufficiency follows by choosing L with $2^L \|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 2B$.

Theorem 3.1 (Full-depth amplification). *Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t + 1\}$ contains an m such that $K(md, N, md)$ holds. Consequently*

$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

Proof. Set $F_t = \Delta_{td}(N)$. Equation (1) yields $F_{t+1} \equiv 2^d F_t + F_1 \pmod{\mathbb{Z}}$. If $K(td, N, td)$ fails, (2) gives

$$\|F_t\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon_t, \quad \epsilon_t = 2(N + 2td + 2)2^{-td}.$$

Two adjacent failures would imply $\|F_1\| \leq 2^d \epsilon_t + \epsilon_{t+1}$, impossible for all sufficiently large t because the left side is positive and the right side tends to zero. Conversely, if F_1 is integral, then every F_t is integral by the same recurrence, so no certificate exists. Finally, (1) is nonintegral for every d, N when S is irrational. If $S = a/(2^c v)$ with v odd, take $N = c$ and $d = \varphi(v)$ to make it integral. $\square$

For instance $D_{1,12,16} = -143140$, with residue 53468 modulo 2^{16} and $B = 31$, supplies a seed. The amplifier produces infinitely many full-depth certificates from this finite fact; it does not provide seeds for all d, N .

3.2 A single canonical endpoint

For $c \geq 0$, odd $v \geq 1$ and a positive multiple H of $\varphi(v)$, put

$$M = \frac{2^H - 1}{v}, \quad B_{H,c} = \sum_{j=0}^{H-1} \varphi(c + 1 + j) 2^{H-1-j}, \quad K = c + H + 1.$$

Euler's theorem makes M integral.

Equivalent formulation 3.2 (Canonical Mersenne gap). The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies

$$K < (-B_{H,c}) \bmod M < M - K. \quad (3)$$

Proof. The identity

$$B_{H,c} = 2^H R_c - R_{c+H} = M(vR_c) - \Delta_H(c)$$

is exact. For $H \geq 1$, $|\Delta_H(c)| < c + H + 1 = K$. Indeed $R_N \leq N + 1$ for $N \geq 1$ and $0 < R_0 \leq 3/2$; the strict inequalities follow by subtracting a positive tail. If $S = a/(2^c v)$, then $vR_c \in \mathbb{Z}$ and $\Delta_H(c) \in \mathbb{Z}$. The residue of $-B_{H,c}$ is therefore within distance less than K of $0 \pmod{M}$, so (3) fails.

If S is irrational, fix c, v . Then vR_c is nonintegral. As H tends to infinity through positive multiples of $\varphi(v)$, $\Delta_H(c)/M \rightarrow 0$ and $K/M \rightarrow 0$. Hence the fractional part of $-B_{H,c}/M = -vR_c + \Delta_H(c)/M$ stays away from both endpoints of $[0, 1]$, which gives (3) for all sufficiently large such H . $\square$

This proof identifies the equivalent missing quantifier. It does not supply H arithmetically without assuming the desired irrationality. The equivalence, with the modulus written as the natural quotient $(2^H - 1)/v$ and v odd, is [Lean source](#). The finite residues also satisfy

$$\rho_{H,N+1,M} = (2\rho_{H,N,M} - \varphi(N+H+1) + \varphi(N+1)) \pmod{M}, \quad \rho_{H,N,M} = (-B_{H,N}) \pmod{M},$$

which is an exact recurrence for searches directed at (3).

4 Information that does not force the gap

The long record supplies a rational control with $c(n) = \varphi(n)$ for odd n , $|c(n) - \varphi(n)| \leq 2$ for even n , $0 \leq c(n) \leq n$, and

$$\sum_{n \geq 1} c(n)2^{-n} = \frac{5}{4}.$$

It has an integral tempered carry with dyadic rank at least $2^e - 1$ at every level. Thus the same lower bound for a hypothetical totient carry is compatible with rationality. For one fixed prime p , the exact laws $g(pn) = pg(n)$ when $p \mid n$ and $g(pn) = (p-1)g(n)$ otherwise, together with $g - \varphi = o(n)$, force $g = \varphi$: a nonzero defect keeps a nonzero ratio to the argument along $p^j n$. The control fails these laws; its full coefficient rank is asserted only at measured finite depths.

A different obstruction concerns rational approximation. Put

$$\Theta_2 = \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}, \quad Q(e, Y) = \frac{(\sum_{d=1}^Y \mu(d)/(2^d - 1)^{e+2})^2}{\sum_{d=1}^Y \mu(d)/(2^d - 1)^{2e+2}}.$$

The divisor-convolution identity gives the displayed value of Θ_2 , as recorded by Fan [6]. The following exact extremum is stated in the long record as Theorem thm:rankonefloor; its proof is [the sharp-floor argument](#) in the linked source.

Theorem 4.1 (Positive rank-one floor). *For $e \geq 1$ and $Y \geq 4$, the denominator of $Q(e, Y)$ is positive, and the unique minimiser is $(e, Y) = (1, 5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1, 5)$.

This theorem treats averages after the quotients are formed. Signed coefficients, coupling before the quotient, and higher-rank kernels require separate arguments. For any proposed rational approximants p_j/q_j in lowest terms, a sufficient arithmetic target is

$$0 < q_j |\Theta_2 - p_j/q_j| \rightarrow 0.$$

Ordinary convergence and Hankel nonvanishing do not contain this denominator estimate. The canonical gap (3) remains the equivalent endpoint; the approximation estimate is one stronger sufficient route.

5 Antecedents and proof record

Allouche and Shallit introduced k -regular sequences [1]. Coons proved that φ is not k -regular [2]; Martin's stronger affine ordering theorem implies the independence in Section 1.1 [3]. The basis theorem gives the exact finite-level normal form. Wong [4] proved least-residue irrationality when the denominator base equals the modulus; Section 2 fixes base two and classifies dyadic observables. The original question appears in Erdős and Graham [5, p. 61].

The exact all-base basis and rank, residue-series irrationality, full-depth amplification, canonical equivalence and positive rank-one floor have proofs in the supplied Lean source snapshot: [basis](#), [residues](#), [amplification](#), [canonical gap](#), and [rank-one floor](#). The historical source links retain 32545d77c4b5; the all-base module requires a matching public source supplement. The integral normal form has its ordinary proof above. The periodic transport lemma is checked in `PeriodicTotientIndependence.lean`; the eventual version follows by the displayed threshold shift. The rational control and the termwise-window and prime-dilation boundaries are proved in `ParityPerturbedRationalControl.lean` and `PrefixValuationAndControlRigidity.lean`, respectively. Finite evaluation ranks and the all-level carry lower bound remain distinct. The long record retains the separate continued-fraction bounds $q \geq 2^{39989}$ and $q > 10^{12038}$ under rationality, alongside the Lean-checked Farey exclusion. Neither implies the universal gap.

References

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